

DECENTRALIZED CRYPTO WORLD BANK

A Blockchain-Based Banking Platform with AI-Enhanced Security and Assistance

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MOTIVATION

1.4B

adults remain unbanked worldwide

\$4.5T

annual MSME financing gap

\$45

average cost per cross-border transaction

6.49%

avg. remittance fee — 2× the UN SDG 3% target

Development finance moves through layered institutions — World Bank, national and local development banks — but settlement is slow, reserves are opaque, and idle capital sits in pre-funded correspondent accounts.

Flat DeFi protocols solve part of this at scale — Aave v3 alone holds **\$26.3B** TVL — but ignore institutional hierarchy entirely:

- One pool for all users — no tiers, no role-based governance
- No cross-tier capital flow between banks and clients
- No behavioral fraud detection or explainable credit scoring
- No on-chain group / solidarity lending for the unbanked

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OBJECTIVES

- Specify a four-tier lending architecture on an EVM chain, testnet-deployed end to end.
- Extend beyond pool lending — deposits, savings, solidarity group loans.
- Add an off-chain ML layer for fraud & anomaly detection with explainability.
- Enforce programmable borrowing limits and role-based access via smart contracts.
- Deploy and evaluate on Polygon zkEVM Cardona / Ethereum Sepolia testnets.

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SYSTEM ARCHITECTURE

Four-tier capital hierarchy — mirrors real-world development finance on-chain.



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TECHNICAL STACK

Presentation

React · TypeScript · WalletConnect

Application Services

Express.js · FastAPI · MCP agent server

Persistence

PostgreSQL · Redis cache

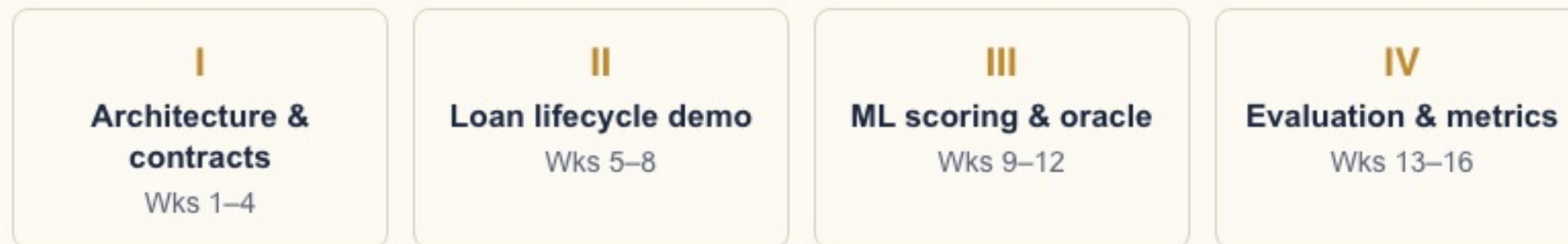
On-Chain Ledger

Solidity 0.8.20 · LoanController

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METHODOLOGY

Agile/Scrum, 2 developers, weekly sync — four incremental phases across a 16-week window.



Solidity React FastAPI PostgreSQL scikit-learn Hardhat

42 Hardhat contract tests passing across four suites — World / National / Local Bank + hierarchy.

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AI/ML FRAUD DETECTION

Random Forest + Isolation Forest (80 trees each) on BCCC-DeFiFraudTrans-2025 (York Univ.), 50k-tx sample, held-out test n = 5,140.



Confusion matrix
on held-out test set
(fraud rate ≈ 8.7%)



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ECONOMIC FEASIBILITY

Gas cost per loan lifecycle (~10–15 txs):



3-yr NPV @ 10% discount

\$149.7k

Projected on \$0 zero-cost prototype build; ROI computed over Years 0–3 with rising benefit-to-cost ratio as tiers scale.

CONCLUSION

CWB specifies a tier-based on-chain banking system that mirrors real institutional finance — World Bank, national and local banks, and clients — instead of the flat single-pool model used by existing DeFi. Combining programmable hierarchy with an AI/ML risk layer and optional conversational agent is, to our knowledge, not addressed together by any prior protocol.

Future work: Polygon zkEVM mainnet-track deployment, InterBankLendingPool & upward deposits, multisig World Bank governance, zkKYC, and Certora formal verification.

HOW CWB COMPARES

DIMENSION	AAVE / COMPOUND	MAPLE / GOLDFINCH	CWB
Architecture	Flat single pool	Single-tier institutional	Four-tier hierarchy
Risk scoring	Collateral ratio only	Manual underwriting	RF + IF + SHAP (F1 0.891)
Governance	Token-weighted vote	Pool-level only	Role-based, per tier
Group lending	Not supported	Not supported	On-chain solidarity groups



github.com/Yuvraajrahman/
Crypto-World-Bank